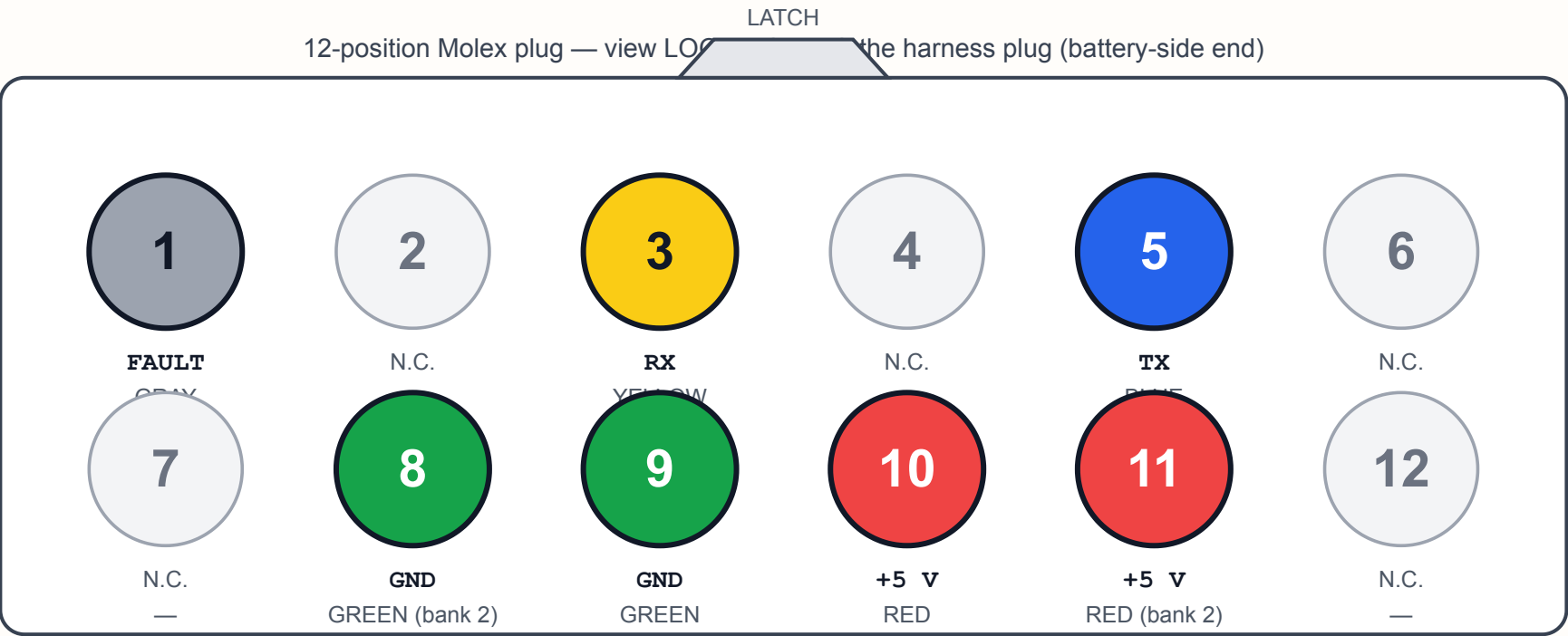


EVTV Tesla BMS harness — controller-end plug pinout

Molex 12-position housing • signals derived from EVTV ESP32 BMS Controller Manual v2.18 (pp. 39, 42)



VIEW: LOOKING INTO the harness plug — the end that connects to the batteries.
Verified with a continuity meter against a real EVTV harness.
The mating controller-side plug is the horizontal mirror of this view.

HOW TO WIRE THE ARDUINO DUE

PIN	SIGNAL	CONNECT TO	WIRE
10	+5 V	Due +5V header	RED
11	+5 V	same +5V rail (jumper to pin 10)	RED
9	GND	Due GND	GREEN
8	GND	same GND rail (jumper to pin 9)	GREEN
5	TX	U2 shifter HV1 (from Due TX1)	BLUE
3	RX	U2 shifter HV2 (to Due RX1)	YELLOW
1	FAULT	leave unconnected (or to a Due input, optional)	GRAY

SIGNAL NOTES

- TX (pin 5) is data OUT of the controller — drives the BMB chain input.
- RX (pin 3) is data IN to the controller — response from the BMB chain.
- Pins 10/11 are both +5 V and pins 8/9 are both GND; tie them together at the Due end. Bank 2 is only populated on ≥4-port harnesses, but bridging is harmless on a 2-port harness (pins 8 and 11 are simply unused there).
- FAULT (pin 1) is a loop that opens if any BMB trips. Optional to monitor.
- NC pins (2, 4, 6, 7, 12) carry no signal in EVTV's harness wiring.
- This is a 6-port fully-connected harness build — no J2 loopback needed.
- The harness plug itself is a Molex 12-position housing. Verify the exact Molex family (Mini-Fit Jr style) against your parts before ordering mating pins or a spare housing.